Gaps and doublets, rational learning, and a knowledge norm on forms*

Roni Katzir
rkatzir@tauex.tau.ac.il

May 2, 2026

Abstract

Inflectional gaps (**forgoed*/**forwent*) and doublets (✓*dived*/✓*dove*) can seem surprising in light of common assumptions about morphology and learning. Perhaps understandably, morphologists have troubled themselves with such cases (especially with gaps) and have offered different ways in which they can be accommodated within the morphological component of the grammar, often by making major departures from common assumptions about morphology and learning. I will suggest that these worries and the proposed remedies are premature. The worries arise from the unmotivated assumption that all gaps and doublets are necessarily derived within individual grammars. Once this assumption is abandoned, the observed properties of gaps and doublets become much less puzzling. The only new assumption needed is that speakers must be able to tell (rather than just guess) that the forms they use are correct.

1 Introduction

In morphological inflection it is common to find exactly one possibility for a given combination of features: *You* ✓*rang*/**rought* but *You* **brang*/✓*brought*. Cases in which more

*Many thanks to Matan Abudy, Adam Albright, Artemis Alexiadou, Moysh Bar-Lev, Jonathan Bobaljik, Anna Bondaruk, Pavel Caha, Barbara Citko, Milica Denić, Danny Fox, Alma Frischoff, Nina Haslinger, Yuval Katz, Eyal Marco, Ora Matushansky, Fereshteh Modarresi, Katya Pertsova, Asia Pietraszko, Adam Przepiorkowski, Ezer Rasin, Adi Raz, Uli Sauerland, Giorgio Sbardolini, Viola Schmitt, Benjamin Spector, Donca Steriade, Yasu Sudo, Jakub Szymanik, Orr Well, Ken Wexler, and Dmitrii Zelenskii, as well as the audiences at TAU, MIT, Leibniz ZAS, MUNI Brno, and the UNPAG Workshop at Göttingen. This work has been supported by ISF 1083/23 and by the Alexander von Humboldt Foundation.

than one possibility is available (doublets; ✓*dived*/✓*dove*, ✓*shrank*/✓*shrunk*) or, perhaps worse, none (gaps; **forgoed*/**forwent*, *have* **strided*/**stridden*) are somewhat unusual.[1] Such cases (for reasons touched upon below, with more emphasis on gaps) have received attention in the generative morphological literature as early as Halle (1973) and Hetzron (1975) and remain an active topic of research today (see Sims 2023 and Pertsova forthcoming and references therein).

The interest in such cases is understandable: while gaps and doublets can in principle be represented on common assumptions about morphology and acquired on common assumptions about learning, the actual properties of gaps and doublets raise several worries. I describe two key assumptions about representations and learning in section 2 and review three worries in section 3, followed in section 4 by some remedies that have been proposed in the literature. In section 5 I explain why I think that the worries are premature and why the remedies should wait until there are good reasons to reject what I take to be a natural (and fairly common) perspective on linguistic knowledge, namely that: (a) the hypotheses that learners entertain need not be monolingual and can in principle include multiple grammars that might not have external differentiating factors such as register (cf. Kroch 1994, Roeper 1999, and Yang 2000); and (b) learners sometimes do not encounter enough evidence to know what the correct hypothesis says about a given form. With multiple grammars, as in (a), otherwise puzzling doublets are predicted. And with incomplete learning, as in (b), gaps follow as long as speakers are required to be able to tell (rather than just guess) that the forms they use are derived within the correct hypothesis. The precise requirement that (b) involves raises complex questions, a matter that I return to in section 5.2. To simplify the presentation, and in analogy with the so-called knowledge norm of assertion (Williamson 1996), I will refer to the requirement in (b) as a knowledge norm on forms:

(1) KNOWLEDGE NORM ON FORMS (KNF): to use a given form x, the speaker needs to know that x is derivable according to the correct hypothesis

[1] The literature notes that gaps and doublets are subject to speaker variation and that even for speakers for whom a given paradigm cell is gapped, one choice might be better than the other. Pinker (1998, pp. 227–228), who notes the *forgo* gap (attributing it to Jane Grimshaw), along with slang *dig* and specific uses of *bear* (to endure) and *become* (to be suitable), marks several different levels of degradation of the relevant forms. To keep notation simple, I use * to mark that at least some speakers find the relevant form degraded. I return to the nature of the degradedness and its source in sections 3.1 and 5.4.1 below.

2 Background assumptions on representations and learning

This section reviews common assumptions about morphology and learning, focusing on their implications for gaps and doublets. Section 2.1 reviews the idea that morphology involves competition for specificity. Section 2.2 reviews the idea that learning involves balancing the simplicity of hypotheses with their fit to the data. These assumptions, which I take to be part of the null hypothesis and will adopt in what follows, make it possible in principle to state and learn gaps and doublets in certain cases (while raising questions about the actual properties of gaps and doublets, as mentioned in the introduction and discussed in greater detail in the following sections).

2.1 Representations: competition for specificity

A simple representational account of gaps and doublets treats the former as an absence of an entry for a given cell in a paradigm and the latter as the existence of multiple such entries. For example, one can imagine that the entries for cells in paradigms are stored individually and that for the past tense of *forgo* there is no lexicalized entry while for that of *dive* there are two.

An immediate objection to this simple idea is that morphology is productive, as seen by *wug* tests (Berko 1958; see Bybee and Moder 1983, Prasada and Pinker (1993), and many others). It is not memorized cell by cell but rather relies on processes of different levels of generality that compete with each other. Even if an entry is not memorized, there is often a general process that can apply and fill the gap. And if there are multiple potential entries, one might expect the most specific one to win and block the others, which in turn should rule out doublets. Below I briefly summarize the idea of competition and why it does not, in fact, rule out gaps and doublets in principle (though, as mentioned, the actual properties of gaps and doublets will still require an account).

Much of morphological theory assumes the Pāṇinian principle of competition for specificity: if two forms match a given context and one is more specific than the other, the more specific one wins. Within Distributed Morphology (DM; Halle and Marantz 1993, 1994, and see Bobaljik 2017 for further discussion and references), which I assume here for concreteness, this competition is referred to as the Subset Principle (see Halle 1997), which says that of the forms that match a given feature specification, the one that matches the most features wins.[2,3] For the present tense in English, for example, *-s* (specified for [present, 3sg]) wins over $\emptyset$ (specified just for [present]). Specificity is generally taken to apply not just to features but also to the context of insertion. For the past tense in English, $\emptyset$ (specified for contexts such as $\sqrt{\text{PUT}}$) and ɪ→ae ablaut (specified for contexts such as $\sqrt{\text{SING}}$) win over the default *-ed*.

As mentioned above, the literature shows that competition for specificity is compatible with gaps and doublets in principle, appearances to the contrary notwithstanding. Gaps can arise from the absence of a default. Consider, for example, the genitive singular form of masculine nouns in Polish, for which Dąbrowska (2001) argues that there is no default. Some forms take the ending *-a*, some forms take *-u*, but neither ending extends automatically to new forms. Kottum (1981) and Gorman and Yang (2019) report that Polish speakers are sometimes uncomfortable with either *-a* or *-u*.[4] For example, Kottum (1981) reports that the speakers he consulted with were unsure about the genitive singular of the city name *Tarnobrzeg*, and Gorman and Yang 2019 identify *balon* 'balloon', among several other nouns, as gapped. These observations can be accounted for if for the relevant speakers, neither *-a* nor *-u* is specified for insertion in the context under consideration (e.g., in the context of $\sqrt{\text{BALON}}$ for speakers for whom *balon* is gapped).

In English, the past tense form *-ed* is clearly a default, as seen by *wug*-testing (Berko

[2]See Embick and Marantz 2008, Embick et al. 2023, and Caha forthcoming for further discussion of competition in DM and in other approaches to morphology. In particular, while Nanosyntax (see Starke 2009, Caha 2009, forthcoming, and Caha et al. 2025) assumes a different architecture and a different notion of competition from DM, the considerations regarding gaps and doublets discussed in the present section are largely the same in both frameworks.

[3]In certain architectures, such as Optimality Theory (Prince and Smolensky 1993), a notion of competition that is very different from that in DM (and relies on an extrinsic ranking of violable constraints rather than on an intrinsic comparison of specificity, among other things) ensures that there will always be at least one winning form. This has been taken to make gaps problematic, but technical solutions have been proposed, both within the framework itself and by embedding the framework within a larger architecture that can rule out winners. The former is advocated by Prince and Smolensky (1993), who propose to include a special candidate output, the *null parse*, along with a constraint MPARSE that allows this candidate to win in certain cases. Crucially, the null parse is not just phonologically empty but ungrammatical, ruling out any structure that includes it. An account of gaps in terms of the null parse is adopted by Raffelsiefen 2004 and Albright 2009, among others. See also Rice (2007) for an OT account couched within McCarthy (2005)'s framework of Optimal Paradigms that allows for the absence of an output without recourse to the null parse. The second solution is advocated by Orgun and Sprouse (1999), who propose to direct the outcome of OT to a second component, CONTROL, that has inviolable constraints. This has been adopted by Fanselow and Féry (2002) and Pertsova (2005), among others.

[4]It is not clear to me to what extent the gaps and doublets reported in the earlier literature hold for current speakers of Polish. Several speakers with whom I consulted did not share the reported judgments. I will set aside the characterization of this variation here and will use the description in the literature for illustration only.

1958) and other considerations, so the account of the *forgo* gap cannot be based on *-ed* not being applicable in the context of $\sqrt{\text{FORGO}}$. But following Arregi and Nevins (2014)'s account of gaps (building, in turn, on Harley 2014), one can say that the expression of the root $\sqrt{\text{FORGO}}$ itself is specified to appear in certain contexts and that these contexts do not include the past tense.

Doublets, meanwhile, can arise from a tie between winners (see Embick and Marantz 2008). English *shrank* and *shrunk* involve ablaut processes that are more specific than the default *-ed* and therefore block it; but neither of these ablaut processes is more specific than the other, so both are possible. Returning to genitive singular masculine nouns in Polish, Kottum (1981) reported that *portfel* 'wallet' can take either *-a* or *-u*, which can be accounted for if both options are specified for insertion in the context of $\sqrt{\text{PORTFEL}}$.

In the English case of *dived/dove*, the fact that *-ed* is a default again complicates things. If both the default *-ed* and the more specific ablaut rule aɪ → oʊ that applies only to certain roots (including $\sqrt{\text{DIVE}}$ but not, e.g., $\sqrt{\text{ARRIVE}}$) are compatible with $\sqrt{\text{DIVE}}$, the latter should win and rule out the former. But again this problem can be overcome.[5] As a first step, one could consider a hyper-specific form of *-ed* that realizes the past tense in the context of $\sqrt{\text{DIVE}}$. This first step is not enough, since this form of *-ed* will be more specific than the ablaut realization (which applies also in the context of roots such as $\sqrt{\text{DRIVE}}$ and $\sqrt{\text{STRIVE}}$). But if we now state a hyper-specific ablaut process that applies only in the context of $\sqrt{\text{DIVE}}$, neither of the two hyper-specific forms will be more specific than the other, and the result will be free variation.

2.2 Learning: balanced simplicity

So morphological theory predicts that certain configurations will result in gaps and that certain configurations result in doublets. Can these configurations be learned?

To answer this question we should say more about how learning occurs. A very general perspective on learning, adopted here and elaborated on below, sees the child as a scientist, trying to find the best hypothesis within a space of possible hypotheses in light of the observed data. This view was central to early work in Generative Grammar (Chomsky 1957, 1965) and will be assumed here (noting that there are other perspectives on learning as well). Within this view, we should say more about: (a) the hypothesis space, (b) the data, and (c) what makes one hypothesis better than another.[6]

Regarding (a), a given morphological theory amounts to a commitment with respect

[5]Though I am not aware of actual proposals that endorse this solution, and below I will mention reasons not to adopt it.

[6]This is not an exhaustive list; e.g., the details of how the child traverses the hypothesis space could also affect what is learned from a given input.

to the hypothesis space: each hypothesis includes one grammar or more (among other things), and morphological theory constrains what these grammars can be. The precise commitment changes between theories. In DM, it says among other things that the grammars include several lists, including one for vocabulary items (VIs), with phonological exponents for insertion in particular contexts. The commitment includes various aspects of how these VIs are written. It also says that VIs are inserted after syntax, subject to competition for specificity.

For (b), a common assumption is that the child relies primarily on distributional evidence: children observe the forms that are used around them and various aspects of the contexts in which these forms are used, but they cannot rely on systematic corrections, paradigmatic organization, morphological decomposition, or other kinds of help. See Marcus (1993) and Calamaro and Jarosz (2015), among others, for discussion.

As to (c), we need to specify the evaluation metric that tells the child how two hypotheses compare in light of the data. The literature highlights the roles of two factors in comparing hypotheses: the simplicity of the hypotheses, sometimes referred to as *economy*, and the tightness of fit of the hypotheses to the data, sometimes referred to as *restrictiveness*. I will discuss each in turn before concluding (with much previous literature) that both factors are crucial and that they should be balanced against each other.

Economy was the evaluation metric of early generative grammar, e.g., as in Chomsky and Halle (1968). For current purposes it can be stated as follows: if two hypotheses licensed by UG are compatible with the data and one is more complex than the other, reject the more complex one. The appeal of economy is that it leads to generalization: for a simple grammar to be compatible with the data, it cannot simply memorize the data (since memorizing requires a very complex grammar) and needs to generalize.

The problem with economy when used on its own is that it leads to over-generalization. Applied to morphology, it will result in doublets everywhere and gaps nowhere. Consider, e.g., how economy would deal with learning the past tense in English. The past-tense VI $\emptyset$ used for $\sqrt{\text{PUT}}$ in actual English is specific: it is listed for a handful of roots and is impossible elsewhere. But the correct VI for $\emptyset$, with a list of applicable roots, is more complex to write than a default such as *-ed*. A default $\emptyset$ is compatible with the data: the observed forms can all be generated. Economy will therefore prefer the wrong, over-generating grammar where both *-ed* and $\emptyset$ are defaults. And when all VIs of the past tense are defaults, speakers will be free to choose for each instance of any root in the context of the past tense which VI to use. In other words, all forms are doublets, an obviously wrong prediction.[7] Economy alone, then, is inadequate: the existence of non-defaults illustrates the known fact that humans generalize in a much more cautious way than what economy

[7] Similar considerations will ensure that the VIs of roots will never be learned as tense-specific, which rules out the second route mentioned above for deriving gaps.

alone supports.

Caution is the intuition behind restrictiveness, which tells the learner to prefer hypotheses that fit the data as closely as possible. We can state it as follows: if two hypotheses licensed by UG are compatible with the input data, but one fits the input data more tightly than the other, reject the one that provides the looser fit. Restrictiveness will prevent the learner from acquiring a default $\emptyset$ VI for the past tense in English: a hypothesis with a VI that allows $\emptyset$ just in the context of $\sqrt{\text{PUT}}$ and several other roots provides a tighter fit, so the overly general hypothesis loses. In this way, restrictiveness makes crucial use of indirect negative evidence: it treats absence of evidence as evidence of absence. See Dell (1981) and Manzini and Wexler (1987) for further discussion of restrictiveness.

The problem with restrictiveness when used on its own is that it is overly conservative: absence of evidence is not always evidence of absence. Since restrictiveness lacks nuance in its inference from indirect negative evidence, it will lead to a surfeit of gaps. For example, if the child has heard *talk* in the present tense but not yet in the past tense, they will immediately treat the past tense as a gap. This is of course empirically incorrect: humans generalize beyond the data, and *-ed* is readily applied by speakers to nonce forms (see Berko 1958, Prasada and Pinker 1993, and Albright and Hayes 2003). So restrictiveness alone is also inadequate.

But having observed the problems of economy and restrictiveness when either is used in isolation, we can now see how each might be able to restrain the other. Economy alone leads to overgeneralization (e.g., $\emptyset$ would become a default), and restrictiveness alone is overly conservative (e.g., *-ed* would *not* become a default), but a learner that balances the two will generalize (for purposes of economy) while still trying to maintain a good fit to the data (restrictiveness). Learning a specific $\emptyset$ VI for the past tense in English, with a list of allowable roots ($\sqrt{\text{PUT}}, \ldots$) is more complex than a default $\emptyset$, and similarly for the ablaut rule, but the gains in restrictiveness more than justify the costs in economy. Similarly, if a given form is only ever seen in the present tense, a balanced learner might conclude that the past tense for that root is a gap: stating this by making the root tense-specific will be costly in terms of economy, but at some point the gains in restrictiveness might justify this. Fleshing out this kind of balance requires more formal detail than I will provide here. One common way of implementing it is known as Minimum Description Length (MDL; Rissanen 1978), with mathematical foundations by Solomonoff (1964); it is also very close to Bayesian inference. For discussion and examples of MDL in the context of grammar learning see de Marcken (1996), Goldsmith (2001), Goldwater and Johnson (2004), Katzir (2014), Rasin and Katzir (2016), and Rasin et al. (2021), among others.

3 Worries

I just reviewed why both gaps and doublets are in principle compatible with common assumptions about representations and learning. Still, there are good reasons not to leave things at that. In line with the literature, my main focus will be on gaps, where the worries appear more serious, but it will be useful to keep doublets as part of the general context.

3.1 Worry 1: Representations

The representation of gaps as missing entries and of doublets as hyper-specific entries raises worries. For doublets, the worry is that hyper-specific entries make a complete accident of the apparent similarities between the *-ed* suffix in *dived* (a hyper-specific entry, inserted only in the context of $\sqrt{\text{DIVE}}$) and in *arrived* (regular *-ed*, not specific to any particular roots) and between the ablaut rule in *dove* (hyper-specific version, applicable only in the context of $\sqrt{\text{DIVE}}$) and in *drove* (more general version, applying also to roots like $\sqrt{\text{WRITE}}$, $\sqrt{\text{DRIVE}}$, and $\sqrt{\text{RIDE}}$). This can be expected to translate into differences in, e.g., psycholinguistic profiles. I am not aware of work that examines these predictions, so at present these remain no more than strange predictions.

For gaps existing evidence makes the worry more serious. If gaps are reduced to missing entries, forms such as *forgoed* or *forwent* should be ungrammatical. As noted in the literature, though, speakers often react to gaps with discomfort, uncertainty, and embarrassment (see Hetzron 1975, Albright 2009, and others). When asked how they would say the past tense of *forgo*, for example, many speakers don't just say that it doesn't exist. They struggle and are often unsure about whatever answer they end up giving (though the discomfort persists even in cases of clear asymmetries between clashing generalizations; see Hetzron 1975 and Pertsova 2016 for relevant discussion). This reaction is an empirical data point that should be accounted for.

It is worth noting that no similar discomfort arises for grammatical impossibilities in morphological derivation ($*$*goed*) or elsewhere ($*$*Jumped cat the*). Deriving all paradigm gaps as ungrammatical (through lack of default, as mentioned above, or by other means) does not predict speaker discomfort. Rather, it leads us to expect, incorrectly, that gaps will be similar to other cases in which unacceptability arises from ungrammaticality.

One might try to relate speaker discomfort in gaps to the absence of any straightforward good option. For run-of-the-mill ungrammaticality there are good alternatives (in a sense that can perhaps be made precise). For $*$*goed*, for example, there is *went*, and for $*$*Jumped cat the* there is *The cat jumped*. For gaps, on the other hand, it is often the case that no potential form is possible. Speakers who dislike both $*$*forgoed* and $*$*forwent*, for example, cannot resort to *do*-support as a repair: $*$*John did forgo treatment* is unacceptable (see Embick and Marantz 2008 and Embick et al. 2023 for discussion). But lack of acceptable alternatives does not seem to explain speaker discomfort in gaps, given that no similar discomfort arises in cases such as island violations, where all potential choices at the relevant point in the derivation are bad and no simple repair is available (e.g., $*$*I know who you saw __ and John* and $*$*I know you saw who and John*). It also fails to explain why there are certain morphological gaps for which no discomfort arises. For example, the missing singular forms in pluralia tantum nouns (**a scissor*), inherently mass nouns (**a furniture*), or certain idioms (**a jink*), all of which arguably involve genuinely missing entries (see Harley 2014), are judged bad and not uncomfortable. See also Pertsova (2005) on distinguishing between forms that feel awkward and forms that feel impossible. Treating all gaps as cases of missing entries makes them uniformly ungrammatical and misses the discomfort that arises with some of them.

3.2 Worry 2: Learning

The hyper-specific entries discussed earlier for doublets such as *dived/dove* can be learned from positive evidence. However, a relatively high number of instances might be needed to justify storing the relevant entries rather than treating one element of the doublet as noise. This predicts a different learning profile for the hyper-specific *-ed* and ablaut rule for $\sqrt{\text{DIVE}}$ than for their more general counterparts, default *-ed* and the ablaut rule applying to $\sqrt{\text{DRIVE}}$. As with representations, while this prediction seems suspicious, I am not aware of attempts to test it.

For gaps the situation is, again, more problematic. Consider what it would take for a balanced learner to conclude that a given form is gapped due to a lack of default. For example, that *forgo* has no past tense because of a VI for $\sqrt{\text{FORGO}}$ that is specific to non-past environments. Such a VI is more complex than a default VI for $\sqrt{\text{FORGO}}$ that is not specified for tense. For a balanced learner, this means that the tense-specific VI would need to have a sufficient advantage in restrictiveness over the default VI if it is to be acquired. The tense-specific VI is indeed more restrictive than the default one, and over a sufficiently large corpus the gains in restrictiveness will outweigh the extra costs of storing the tense-specific entry. But given that *forgo* is a relatively rare verb to begin with, it will take a very large amount of data for the learner to reach this point. In principle, one can imagine the relevant property of *forgo* being shared with other forms, which could allow this property to be acquired based on the aggregate frequency of these forms. But it is hard to see what this shared property or the other gapped forms might be, especially in light of the fact that seemingly related forms such as *undergo* are not gapped. All of this makes an account of the *forgo* gap in terms of a tense-specific VI for $\sqrt{\text{FORGO}}$ unlikely.

Person	*abolir* sg	*abolir* pl	*dormir* sg	*dormir* pl
1	—	abolimos	duermo	dormimos
2	—	abolís	duermes	dormís
3	—	—	duerme	duermen

Table 1: Spanish defective *abolir* lacks present-tense forms in the singular and in the 3pl, where stress falls on the stem. In these cells, both the diphthongized form (**abuelo*, **abueles*, ...) and the non-diphthongized form (**abolo*, **aboles*, ...) are rejected by speakers. Compare regular diphthongizing *dormir* 'to sleep'.

3.3 Worry 3: Distribution

The outline of representations and learning above says nothing about where gaps and doublets appear. This can make them seem completely arbitrary, in line with Halle (1973)'s proposal that gaps are simply forms that are marked for non-insertion. As discussed by Hetzron (1975), Albright (2003), and others, however, gaps (and doublets) do not seem to be scattered randomly across inflectional paradigms. Rather, they tend to be concentrated in specific paradigmatic corners. Informally speaking, these corners seem to involve the presence of multiple generalizations that could potentially apply to a given item.

Doublets, by definition, involve two different possibilities for the realization of a given form. Very often, each of these possibilities reflects a different generalization, as in the case of *dived* (using the default *-ed*) vs. *dove* (as in *drive*∼*drove* and *strive*∼*strove*).

Clashing generalizations seem relevant also for the distribution of gaps, though here the facts are more complex than for doublets and in some key cases are still contested in the literature. For genitive masculine singular nouns in Polish, the gaps reported in the literature seem related to the availability of two generalizations: add *-a* vs. add *-u*. Similar comments apply also for the present-tense gaps in the Spanish verbal system observed by Harris (1967, pp. 122–3) and discussed by Albright (2003, 2009), Boyé and Cabredo Hofherr (2010), Maiden and O'Neill (2010), Gorman and Yang (2019), and others. These gaps arise in parts of the paradigm where an alternation could apply, and Albright (2009, p. 124) notes that for the gapped verb *abolir* 'to abolish', speakers consider both the alternating form *abuelen* and the nonalternating form *abolen*. See Table 1. The same holds also for the genitive plural gaps in Greek studied by Sims (2006, 2009), which occur in a part of the paradigm for which the placement of stress is unpredictable: for some nouns stress shifts (e.g., *tavernón* for *tavérna* 'tavern'), while for other nouns within the same inflection class it doesn't (e.g., *mitéron* for *mitéra* 'mother'). See Table 2.

The clashing generalizations that accompany a given gap need not be equally good (or bad). In fact, while speakers might volunteer (and reject) the outcomes of applying

Case	*mitéra* sg	*mitéra* pl	*tavérna* sg	*tavérna* pl	*kopéla* sg	*kopéla* pl
Nom	mitéra	mitéres	tavérna	tavérnes	kopéla	kopéles
Gen	mitéras	mitéron	tavérnas	tavernón	kopélas	—
Acc	mitéra	mitéres	tavérna	tavérnes	kopéla	kopéles

Table 2: Modern Greek noun paradigms, from Sims (2023, table 9). The genitive plural can appear with stem stress, as in *mitéra* 'mother' or with suffix stress, as in *tavérna* 'tavern'. For *kopéla* 'girl' it is gapped.

Person	yap- sg	yap- pl
1	yap-ası-m	yap-ası-mız
2	yap-ası-n	yap-ası-nız
3	yap-ası-∅	*yap-ası-ları / *yap-a-ları

Table 3: Turkish *-AsI* desideratives, from İleri and Demirok (2022), lack a 3pl form for most speakers. In this cell, both the form with the full desiderative suffix (**yap-ası-ları*) and the form with the shortened desiderative (**yap-a-ları*) are rejected; individual speakers prefer one over the other but find their preferred form degraded. The 3sg cell (*yap-ası*, 'feels like doing') is itself irregular: the portmanteau *-ası* spells out both DESID and 3SG.POS.

two or more clashing generalizations when asked about a given gap, they might still have a clear sense that one of those outcomes is better than the others. The *forgo* gap illustrates this kind of asymmetry. The relevant generalizations in this case are the default *-ed* and the stem change that applies in *go*∼*went* and *undergo*∼*underwent*. But there is an asymmetry between the two generalizations: speakers generally dislike *forgoed* and prefer *forwent* when forced to choose.[8] Nevertheless, speakers who are confronted with the need to produce the past tense of *forgo* occasionally mention both *forgoed* and *forwent* (before rejecting both), which suggests that both generalizations are live options. İleri and Demirok (2022), who discuss a gap in the 3pl desiderative in Turkish, see Table 3, note that each speaker prefers either of two conceivable forms but rejects even their preferred form.

The case of 1sg non-past gaps in Russian, discussed by Halle (1973), Hetzron (1975), Sims (2006), Baerman (2008), Albright (2009), Pertsova (2016), Gorman and Yang (2019), and others further illustrates the possibility of asymmetries between clashing generalizations. The gapped forms in this case generally belong to the subclass of second conjugation

[8] See Sims (2006, pp. 14–17) for usage statistics that show the skew in favor of *forwent*.

Person	*sprosit'* sg	*sprosit'* pl	*pobedit'* sg	*pobedit'* pl
1	sprošu	sprosim	—	pobedim
2	sprosiš'	sprosite	pobediš'	pobedite
3	sprosit	sprosjat	pobedit	pobedjat

Table 4: Russian *pobedit'* 'to defeat' lacks the 1sg non-past form. In this cell, the two potential palatalized forms — one involving the *d*∼*ž* alternation (**pobežú*) and one involving the *d*∼*žd* alternation (**pobeždú*) — as well as the nonalternating form (**pobedjú*) are all unacceptable; compare regular *sprosit'* 'to ask', which takes *sprošú* in this cell via *s*∼*š*.

verbs with a stem-final dental consonant; see Table 4.[9] This consonant undergoes palatalization in the 1sg non-past. When the consonant is *t*, the palatalization alternation offers two different generalizations that could in principle apply: *t*∼*č*, *t*∼*šč*. When the consonant is *d*, *s*, or *z*, on the other hand, the form of the palatalization seems deterministic — *s*∼*š*, *d*∼*ž*, and *z*∼*ž* — and yet gapped verbs that end in these consonants exist. However, Gorman and Yang (2019) provide evidence (building on observations by Sims 2006 and Slioussar and Kholodilova 2013) that a clashing alternation *d*∼*žd* is considered by speakers also for stems ending in *d*, and that even for stems ending in *s* and *z* (for which no other alternation is applicable), there is a competing generalization: namely, not to alternate at all. Gorman and Yang (2019)'s conclusion is further supported by the experimental results of Pertsova (forthcoming), which show that children and adults produce both alternating and nonalternating forms in the relevant cell for nonce verbs.

But while the correlation between gaps and clashing generalizations does not seem accidental, establishing causality is not straightforward. As the examples discussed above illustrate, there are in principle three different potential outcomes to clashing generalizations, all of them attested:

(2) Three possible outcomes of clashing generalizations:
  a. 0 (gap): neither generalization leads to an acceptable form
  b. 1: one or the other of the clashing generalizations wins over the other, resulting in a single licit choice
  c. 2 (doublet): both options are acceptable, leading to free variation

The representational and learning considerations above do not explain the connection between defectiveness and clashing generalizations, let alone predict which clashes lead to gaps, which to a single winning form, and which to doublets.

[9] For one verb, *zatmit'* 'to eclipse', the stem-final consonant is labial rather than dental. Pertsova (2016) suggests that the reasons for this gap are different from those of the dental-final stems.

4 Responses

The literature includes several ways of enriching the grammar, the learner, or both in an attempt to address one or more of the worries above. Given that the challenges regarding gaps are much clearer than those for doublets, it is not surprising that the literature has paid more attention to the former than to the latter. I will do the same. I discuss several central proposals for gaps, focusing on properties of these proposals that relate to the three worries above and on where these proposals deviate from the null hypothesis.

4.1 Daland et al. (2007): Learning distributions

Daland, Sims, and Pierrehumbert (2007) propose that gaps are paradigm cells with low probability (see Sims 2015 for further discussion and elaboration). This can be seen as a concrete implementation of Halle (1973)'s analysis of gaps through filters on insertion. The grammar itself can in principle include general morphological processes, but some forms are then filtered out. Daland et al. (2007)'s proposal can be taken to provide missing details about representation and learning in Halle's account: representationally, the grammar includes distributions over cells in paradigms, with a threshold on probabilities acting as a filter; and the acquisition of this filter proceeds through balanced learning.

By relying on a filter (specifically, one that maintains distributions over paradigm cells), Daland et al. (2007) enrich the morphological part of the grammar considerably. It is far from clear why such a representation should be maintained and even less clear why it should act as a filter, an instance of the well-known problem of reducing unacceptability to low probability; see Chomsky 1957. And even if one were to add a stipulation that rules out low-probability cells in paradigms, it is hard to see why this would make speakers uncomfortable and not simply judge the relevant forms as bad. That is, the worry about representations, discussed in section 3.1, remains unaddressed.

In order to account for gaps in low-frequency forms and for the skewed distribution of gaps, Daland et al. enrich the learner so as to take into consideration lexical neighborhoods, defined in a particular phonological way. This allows the learner to induce low frequency for cells even for some low-frequency stems, based on low-frequency cells in lexical neighbors. However, it does not explain why *forgo* is gapped in the past tense despite low frequency and no gapped neighbors. So the worry about learning in section 3.2 remains unaddressed.

The distributional worry in section 3.3 also remains unaddressed: nothing in Daland et al. (2007)'s framework explains why gaps should follow clashing generalizations.

4.2 Yang (2016), Gorman and Yang (2019): Productivity

Yang (2016) and Gorman and Yang (2019) relate gaps to a notion of *productivity*. Specifically, gaps are taken to arise for a given cell when (a) no form is observed for that cell, and (b) none of the processes that could potentially fill that cell are productive. Gorman and Yang implement this idea using the *Tolerance Principle* (Yang 2005, 2016), a special learning criterion that determines when a given process is productive: if there are few exceptions to the process, it is productive; otherwise it is not.

A general concern regarding the Tolerance Principle is that it treats productivity as a value that needs to be inferred (a view shared also with the Bayesian approach of O'Donnell 2015). This is a departure from the common morphological perspective discussed in section 2.1, which recognizes no such value. On approaches such as DM, productivity might be used descriptively by the morphologist to convey a sense of how broadly and flexibly a given vocabulary item can be used, but it is an informal, gradient notion that can be read off a given grammar rather than a value that is written into the grammar. And if there is no productivity value that needs to be learned, there is no clear use for a special learning criterion for such a value.

If one goes beyond common morphological assumptions and incorporates a productivity value into the representations, one needs to ask how it interacts with the morphological grammar. Gorman and Yang's discussion suggests that they see the productivity value as a filter: the morphological grammar has processes of different levels of generality, but even when a given process provides a form, this form would be ruled out if the process is marked as non-productive. In this way, the Tolerance Principle serves as another instantiation of Halle (1973)'s filter: forms that are generated by a non-productive process are ruled out unless they have also been lexicalized.

Like other implementations of Halle (1973)'s filter, the Tolerance Principle does not address the representational worry discussed in section 3.1: even if one accepts the filter, it is unclear why violating it should sometimes lead to discomfort. For example, why should the Spanish and Russian gaps mentioned above, which Gorman and Yang analyze as violations of the Tolerance Principle, lead to any more discomfort than the lack of singular **scissor* or **jink*?

Another representational worry concerns gaps with defaults. For *forgo*, for example, the Tolerance Principle needs to say that *-ed* is not productive. But it is not clear what conception of productivity would make *-ed* productive for the forms that it extends to (*google*, *fax*, *wug*) and nonproductive for *forgo*. Gorman and Yang suggest that *forgo* is forced to follow the *go*∼*went* alternation (which is not productive and therefore ruled out) and cannot use *-ed*, but the details of why *-ed* is inapplicable remain unspecified. See Pertsova (forthcoming) for a variant of this problem for the case of 1sg non-past verbs in

Russian.

The Tolerance Principle does offer a handle on the two remaining worries. For learning, what needs to be acquired on the Tolerance Principle account is the unproductivity of a given rule rather than a specific property of individual forms, so low-frequency forms pose no particular challenge in principle. As to clashing generalizations, items for which just one generalization applies can count as exceptions to the other generalization. Depending on the precise counts, the number of exceptions to each generalization might be sufficiently high to prevent it from being productive. This allows the Tolerance Principle to link gaps to clashing generalizations.

4.3 Nevins (2014), Mendes and Nevins (2022), Müller (2020): Lethal competition

Nevins (2014) and Mendes and Nevins (2022) propose that some gaps arise through what they refer to as *lethal competition* between two derivations within a single grammar (see also Andrews 1990, Hudson 2000 and Fanselow and Féry 2002). These gaps are attributed to the presence of clashing morphological expressions within the grammar, where neither expression is more specific than the other (so competition for specificity does not force a single winner). For other gaps, Mendes and Nevins (2022) propose lexicalization through lack of default, following Arregi and Nevins (2014). Lethal competition provides yet another implementation of Halle (1973)'s filter: generated forms are ruled out if a competing derivation applies.

Being a filtering approach, lethal competition does not explain why some gaps lead to speaker discomfort rather than simple unacceptability. More worrying is the fact that ties within the grammar are generally taken to lead to free variation and not to ungrammaticality. Embick and Marantz (2008) discuss this for the possible derivational doublet *cover*-$\emptyset$ and *cover-age*. Similarly, a tie between the ablaut processes deriving *sank* and *sunk* is presumably accountable for the fact that both forms are possible. So accounting for some gaps in terms of lethal competition comes at the cost of the usual account of free variation, in doublets and elsewhere. See Albright (2009) for relevant discussion.

Turning to learning and distribution, lethal competition offers a partial handle on the challenges. The clashing morphological realizations within the grammar represent competing generalizations. It also helps in accounting for the learning of gaps in low-frequency forms: the clashing generalizations are presumably acquired based on a range of different forms, so the low frequency of individual forms is irrelevant. Note, however, that this does not help in the case of *forgo*: since one process is more specific than the other, lethal competition is not applicable. This gap would need to be lexicalized, which leaves unaddressed the challenge of learning such an isolated, low-frequency gap. It also leaves unexplained the distributional connection of gaps such as *forgo* with competing generalizations.

A possible response to the challenge of reconciling lethal competition with free variation is to restrict it to specific cases. This idea is pursued by Müller (2020), who suggests that gaps arise when competing processes formally trap each other: each process introduces a problem that is fixed by the other, leading to an infinite loop. Not every architecture allows for an implementation of this idea, but Müller (2020) provides a system couched within a variant of Harmonic Serialism (McCarthy 2000, 2010) in which such infinite loops can arise.

As far as I can tell, cases such as *forgo* remain unaccounted for. One could devise constraints that would lead to oscillation between *forgoed* and *forwent*, but I do not see how this constraint configuration can be given a principled explanation or how it might be acquired. It is also hard to see how these constraints would not similarly afflict the ungapped *go* and *undergo*. A possible attempt to isolate *forgo* might appeal to *went* and *underwent* being listed, contrasting with the lack of a memorized *forwent*. This, however, predicts that complex nonce verbs with *go* would be gapped. As far as I can tell, however, a nonce *rego* (meaning go again) would have *rewent* as its past form. Similarly, while constraints can be devised to trap unlisted forms in Russian 1sg non-past gaps, which Müller (2020) suggests is the account of these gaps, this would predict that nonce verbs with the same phonological profile will be uniformly gapped in this cell. As Pertsova (forthcoming) shows, this is not the case: some nonce forms are gapped, but others are not.

4.4 Pertsova (2005): Conservatism

The above highlighted the difficulty of making competing generalizations directly responsible for gaps within the grammar without sacrificing doublets and other cases of free variation. Pertsova (2005, 2016, forthcoming) addresses this challenge by ruling out each competing outcome on its own rather than the configuration of competing generalizations as a whole: gaps arise not because both processes try to apply but because neither can. Specifically, she focuses on gaps that involve a potential alternation and proposes that these gaps are cases where (a) the alternation is not supported, but (b) not alternating is not permissible either.

To implement this idea, Pertsova adopts Steriade (1997)'s principle of Lexical Conservatism, which requires that every property of a novel form have support from a listed form. In the case of 1sg non-past form of Russian verbs ending in dental consonants, the relevant property that might need support is the palatalization that is expected to take place in the 1sg form. Pertsova (2016) notes that the past passive participial forms of a verb show the same palatalization alternation. The verb *vozit'* 'to transport', for example, undergoes the same palatalization in the past passive participial forms — e.g., *vozhennij* — and in the 1sg non-past *vozhu*. Not all verbs have such passive participial forms, however, and Pertsova (2016) shows that the absence of such forms is a good predictor for gaps.

Accounting for gaps in terms of Lexical Conservatism raises several concerns. Starting from the representation of gaps, it is unclear why lack of supporting forms might lead to gaps to begin with (see Sims 2017, 2023). Steriade (1997)'s argument for Lexical Conservatism involves cases where lack of support for an alternation from a different form corresponds to nonalternation, but the result is judged acceptable. For example, Steriade (1997) uses Lexical Conservatism to account for stress shift before *-able* in English: *rémedy* becomes *remédiable*, supported by the stress-shifted form *remédial*, while *párody*, for which no stress-shifted form such as **paródial* is available, resists stress shift and becomes *párodiable* rather than **paródiable*. Why should Russian 1sg gaps be different, with lack of a supporting alternating form resulting in unacceptability rather than in nonalternation?

One could attempt to address this worry, as Pertsova does, by making the alternation requirement very strict for the relevant Russian verbs, so that if they alternate they will violate Lexical Conservatism and if they don't they will violate the alternation requirement. This approach, which amounts to two separate filters (one that rules out unsupported alternations and one that rules out nonalternating forms) requires a further mechanism to ensure that when all output possibilities violate requirements that are sufficiently strict, the result is ungrammatical. Pertsova 2016 refers to this as a threshold approach and discusses a way in which it can be implemented within a particular constraint-based system. As far as I can see, a threshold mechanism would serve the sole purpose of ruling out winners within an architecture that otherwise allows winners to surface. In addition to being an otherwise unmotivated complication, a threshold approach faces an empirical challenge: it needs to make each of the forces that rule out potential candidates — in the current case, both Lexical Conservatism and the alternation requirement — sufficiently important to ensure that violating either would prevent a form from passing the threshold. Both parts are necessary for the threshold approach, but neither seems straightforward. Speakers extend alternations to novel forms, including in Spanish (see Albright 2009, pp. 132–133) and Russian (see Tomas et al. 2017 and Pertsova forthcoming). This suggests that Lexical Conservatism, to the extent that it holds, is not sufficiently important to block an output. And the importance of an alternation requirement is challenged by Slioussar and Kholodilova (2013)'s evidence that various recent borrowings from English into Russian surface without alternation in the 1sg non-past and by Pertsova (forthcoming)'s experimental results that show that children and adults sometimes produce nonalternating forms for nonce dental-final verbs in the 1sg non-past.

Even if one provides a way to derive unacceptability (and not just nonalternation) using Lexical Conservatism and a threshold approach, speaker discomfort remains unaddressed:

if alternation and nonalternation are ruled out by the grammar, each should just be judged as unacceptable.

And even if the challenges above are addressed, some gaps remain unexplained. As Pertsova (2016, p. 17) notes, several defective verbs do, in fact, have alternations elsewhere. These gaps would then require a different explanation. Separate explanations would also be required for the gaps discussed for Greek by Sims (2006), where the nonalternating form should be unproblematic and for the Turkish gap discussed by İleri and Demirok (2022), where the matter of alternation seems irrelevant to begin with.

Turning to learning, Lexical Conservatism offers a potential partial handle on the acquisition of gaps: the principle is presumably innate, and it affects low-frequency forms that are candidates for an alternation just as easily as it does high-frequency forms.

As to the distribution of gaps, the account weakens the connection between gaps and clashing generalizations, given that Lexical Conservatism does not require nonalternation to be a grammatical possibility. Pertsova (2016) suggests that this is an advantage of her approach, since in the case of Russian 1sg non-past forms, not alternating does not seem to be a licit option for verbs ending in dental consonants. But the observations of Sims (2006), Slioussar and Kholodilova (2013), Gorman and Yang (2019), and Pertsova (forthcoming) mentioned above support the presence of conflicting processes also for Russian 1sg gaps and strengthen the case for deriving gaps from competing processes. Deriving gaps through a separate elimination of each potential outcome treats the correlation with competing processes as an accident.

One can of course imagine other implementations of the idea that gaps are due to the impossibility of each potential outcome on its own, and the literature occasionally mentions factors other than unsupported alternations that might rule out potential outcomes. Such factors include phonotactic violations and a pressure for homophony avoidance. See Pertsova (2005), Baerman (2011), and Sims (2023), among others. As the literature notes, accounting for gaps using such factors is difficult, leaving the matters of representation, learning, and distribution currently open.

4.5 Albright (2003, 2009): Uncertainty

Albright (2003, 2009) accounts for gaps by incorporating uncertainty into the grammar. Gaps are taken to be cases in which speakers have low confidence in a given trans-derivational connection and no lexicalized form. To implement this idea, Albright (2009) proposes that the grammar represents connections between different parts of different paradigms and that these connections are inferred by learners using a particular method, namely the Minimal Generalization Learner of Albright and Hayes (2002). For low-confidence connections, such as the gap-prone parts of the Spanish and Russian paradigms, a localized form of Lexical Conservatism is implemented: in those mappings, the null parse wins unless there is a listed form (for which Albright uses a constraint LEX that penalizes novel forms). For higher-confidence connections, the null parse does not win regardless of whether a listed form exists.

I believe that speaker uncertainty is indeed central to an account of gaps. In this key regard, the present proposal will be in agreement with Albright (2003, 2009)'s account. But representing trans-derivational dependencies grammatically is a significant architectural commitment, and it also remains unclear why grammar-internal uncertainty should lead to gaps and in particular give rise to speaker discomfort. The first worry, from section 3.1, is therefore unaddressed.

Turning to learning, gaps in low-frequency cases are expected, since low frequency is compatible with low confidence and can in fact contribute to it. The learning method that Albright uses to induce trans-derivational constraints and their corresponding confidence scores is likewise problematic. As noted by Albright (2009, 119–120) and by Pertsova (2016, p. 10), this choice of learning method is not accidental: for the proposal to yield gaps, the learner needs to ignore broad, robust generalizations. For example, deriving the gaps in the past participle of *stride* requires generalizing narrowly based on the simple past form and ignoring the broad generalization that *-ed* is the participial ending in the great majority of cases. This is at odds with the general framework of balanced learning, discussed in section 2.2, leaving open the challenge of acquiring full morpho-phonological grammars from distributional evidence.

Empirically, Pertsova (2016, pp. 11–14) notes that the specific learning approach used by Albright (2003, 2009) does not seem to make the right predictions for the case of Russian 1sg non-past gaps.

Finally, clashing generalizations are expected to be relevant, since they serve to increase uncertainty. However, it is also worth noting that Albright's proposal does not extend to clashing generalizations in cases such as *forgo*: even if one accepts the directional dependencies that build, e.g., the past participle from the simple past, there is no similar basis for deriving the simple past that could make *forgo* problematic. Similarly, as far as I can tell, for the genitive plural in Greek discussed by Sims (2006) and for the Turkish desiderative discussed by İleri and Demirok (2022).

In what follows I will keep Albright (2009)'s idea that gaps are cases of uncertainty, but I will treat the uncertainty to be about the correct hypothesis rather than being within the grammar. I will show that this idea fits naturally within common views on representation and learning.

5 Proposal

As reviewed in the previous section, different proposals in the literature take gaps and doublets to require nontrivial departures from the null hypothesis. Such departures might be merited if the empirical gains from these departures are sufficiently significant to justify the costs incurred in complicating the theory. As I tried to show, this is not the case for the proposals reviewed: the theoretical complications are problematic not just conceptually but also empirically. In particular, none of the existing proposals adequately addresses the three worries regarding representations, learning, and distribution of gaps.

Common to the proposals reviewed in the previous section is the assumption that gaps and doublets should be derived within individual grammars. This assumption is unwarranted, and the present section shows that once this assumption is abandoned, gaps and doublets become much less puzzling. Abandoning this assumption will require talking about the broader context in which grammatical knowledge is situated. For presentational convenience, and in line with the view of the child as scientist, I refer to this broader context as *hypotheses*, which I use as terminological shorthand for internal representations of certain aspects of reality, including grammars among many other things. This terminological choice is not meant to imply in any way that there is a psychological reality to hypotheses as a unified whole beyond that of their individual representational parts. I will mention some properties of these representational parts and of speakers' attitudes towards them that I believe are relevant for gaps and doublets, but this will have to be brief and tentative. In particular, I will remain agnostic about the representation of aspects of external and internal reality beyond grammar.

5.1 Assumptions about hypotheses

In rejecting the requirement that gaps and doublets be derived within individual grammars I will rely on two assumptions about hypotheses discussed in the literature and mentioned briefly in the introduction:

(3) Assumptions about hypotheses

  a. MULTIPLE GRAMMARS WITHIN A HYPOTHESIS: a hypothesis can include multiple grammars, which may have specific contexts of use (register) but may also be in free variation

  b. UNCERTAINTY ABOUT THE CORRECT HYPOTHESIS: acquisition is sometimes incomplete, in the sense of leaving human learners with uncertainty about various aspects of the correct hypothesis

Assumption (3a) is supported by bilingualism and register and has been argued for by Kroch (1994), Roeper (1999), Yang (2000), and others. It extends the discussion of representations from section 2.1: each of the (potentially many) grammars represented in a given hypothesis has a morphological component in which competition for specificity takes place; but there is no competition for specificity between forms that belong to different grammars within a hypothesis. This lack of competition for specificity between grammars will allow for the derivation of doublets without recourse to hyper-specific entries, even if individual grammars do involve competition for specificity.

As far as I can see, (3a) is not derivable from first principles. One can imagine beings otherwise similar to us that are inherently monolingual. Still, in light of a range of familiar observations, it is clear that actual humans have the capacity to represent multiple grammars.

Assumption (3b) will allow for the derivation of gaps. This assumption is at the heart of the view of humans (both children and adults) as scientists. On this view, humans behave as if there is a correct hypothesis and as if they can come closer to this correct hypothesis through inquiry: they collect evidence and use it to reason about different parts of their internal representations. In doing so, they can have different degrees of confidence in different representational parts. They can treat one part as an established fact while remaining uncertain about another.

Like the representational claim about multiple grammars, the claim about humans as scientists is substantive. For example, one can imagine beings who (as children, as adults, or both) learn to improve their predictions about their environment but do so without reasoning about representational parts of hypotheses. One can also imagine beings who do reason about representational parts of hypotheses but do not maintain a record of their confidence in the different representational parts. For example, they might simply adopt the complete hypothesis that is best in terms of its balance of simplicity and fit. But as with the representational claims above, I believe that the view of the child and the scientist has sufficient merit in light of known observations in the literature to be adopted here.

Before showing how the assumptions in (3) support an account of gaps and doublets, I will need to be more explicit about the effects of the uncertainty in (3b).

5.2 A condition on the use of forms

How does the epistemic assumption in (3b) relate to judgment and action? One could imagine, for example, that in the face of uncertainty about what the correct hypothesis says, speakers would accept all leading possibilities, perhaps sampling from them in proportion to their probability and judging them accordingly. One could also imagine that speakers would always opt for the forms favored by the leading hypothesis. Neither of these choices would derive gaps, however, and in section 1 I suggested a third choice, according to which using a form requires being able to tell (rather than just guess) that it is grammatical according to the correct hypothesis. Descriptively, and using 'know' loosely, the condition can be stated as follows:

(4) KNOWLEDGE NORM ON FORMS (KNF; repeated from (1)): to use a given form x, the speaker needs to know that x is derivable according to the correct hypothesis

KNF is modeled after the knowledge norm of assertion (KNA), defended by Williamson (1996, 2000) and discussed in much subsequent literature; see Unger (1975), DeRose (2002), Turri (2016), and Benton (2024), among many others. In the domain of meaning, it has been argued that asserting something requires knowing that it is true and not just believing that it is likely. If I haven't checked the weather outside I can neither assert that it's raining nor that it's not raining, even if I have a strong probabilistic belief for one of the options (e.g., if yesterday's weather forecast said that there is an 80% chance that it would rain at this time today). If forced to assert one or the other, I would presumably choose the likelier option, but the assertion would still be infelicitous. The current proposal, then, says that not just the content of assertions but also the forms used in utterances are subject to a requirement of knowledge.

For assertion, the precise requirement has been a matter of ongoing debate. See, e.g., Douven (2006), Lackey (2007), Mandelkern and Dorst (2022), and Graham and Pedersen (2024) for arguments that assertion is weaker than KNA suggests and can be justified by less than actual knowledge. What seems much closer to a consensus is that the requirement is strong and amounts to more than just a plausible guess, as illustrated by the weather example above among many other considerations.[10] It is possible that the term 'knowledge' is misleading in this context. Perhaps it suffices that an assertion adequately reflects what we treat as established facts within the correct hypothesis. Be that as it may, I will assume that the strength of the requirement for KNF is the same as for KNA (and perhaps also for norms of use in other areas of rule-based behavior).[11]

KNF is no more stipulative than other choices that could be made, such as license to use forms that are grammatical according to some high-ranking hypothesis or license to use forms that are grammatical according to the highest-ranking hypothesis. And in light of the broader context that includes KNA, KNF strikes me as more natural than those other theoretical choices. But I don't know if this choice might reduce to more basic considerations.

[10] An exception here is Mandelkern and Dorst (2022), who suggest that a plausible guess can suffice for assertion. See, however, van Elswyk and Benton (2023) for a critical discussion.

[11] Another question is why speakers should be bound by conditions such as KNA and KNF in the first place.

5.3 Deriving gaps and doublets

Let us see how the null hypothesis and KNF can derive gaps and doublets, starting from the latter and from assumption (3a). Adult speakers have at least one grammar, but typically many more. This is most clearly the case for multilingualism but can also be seen in the ability of speakers to switch between registers. More broadly, as discussed by Roeper (1999) and others, speakers can be seen as having many different grammars as part of the adult state. For acquisition, then, the hypotheses that the child considers can encode *sets* of grammars rather than being limited to just a single grammar. It is often the case that all the grammars in the winning hypothesis will agree. For example, the different grammars in the winning hypothesis of English speakers will agree that the past tense of *walk* is *walked*. But it is conceivable that the winning hypothesis will contain two grammars that disagree about a given form. For example, $G1$ might derive *dived* (by not having an ablaut rule that applies to *dive*, so only the default *-ed* will be applicable) and $G2$ might derive *dove* (by having such an ablaut rule, which will win the competition for specificity). In this case, and unless the grammars differ in when they can be used (as in the case of register), both forms will be possible. Note that while there is competition between (the processes behind) *dived* and *dove* within $G2$ (with *dove* as the sole winner), there is no such competition within $G1$ (where *dove* is simply not derived). The observed variation between *dived* and *dove* arises not within either of the grammars but from the multilingual configuration. Of course, encoding both $G1$ and $G2$ in H is more complex than having just one, which on common assumptions about simplicity (as mentioned above) requires more evidence for acquiring H than for acquiring a hypothesis that has just $G1$, for example, or just $G2$. Note, however, that since the differences between $G1$ and $G2$ are likely to be very minor, a sensible representation of $H = \{G1, G2\}$ will take advantage of this overlap and not be much longer than either $H1 = \{G1\}$ or $H2 = \{G2\}$.[12] For example, the representation can include diacritics for $G1$ or $G2$ that are used only when an element of the grammar belongs to just one of the two grammars. And encountering both *dived* and *dove* at above-noise levels in the primary linguistic data should suffice for a rational learner to justify adopting H (rather than $H1$ or $H2$).[13] This provides an account of doublets involving a more specific and a less specific process, even within a framework that involves competition for specificity.[14]

[12]To simplify the discussion I write things like $H = \{G1, G2\}$, setting aside everything in a hypothesis other than the grammar or grammars within it.

[13]For doublets where neither process is more specific than the other, such as *shrank* and *shrunk*, representation within an individual grammar is still possible. And since it is the representationally simpler option, it will be acquired, and no split will arise.

[14]Multiple grammars can also account for doublets in frameworks, such as OT, where there is no inherent competition for specificity but where the basic architecture limits the derivation of ties. See Anttila (2004,

Turning to assumption (3b) — and to gaps — consider what happens when the learner is uncertain about a given aspect of the correct hypothesis. For example, the learner might not be sure what past tense is licensed for *forgo* by the correct hypothesis: perhaps they find it equally likely that the correct hypothesis licenses *forgoed* and that it licenses *forwent*, or perhaps they find one of the possibilities much likelier than the other but are still not certain.

Why should speakers be uncertain? One possibility is that learners entertain the hypothesis of a VI for $\sqrt{\text{FORGO}}$ in the past tense that involves stem suppletion analogously to but separate from $\sqrt{\text{GO}}$. A more plausible possibility, pointed out to me separately by Adam Albright and Jonathan Bobaljik, is that the uncertainty is about whether *forgo* is decomposed into *for+go* or not: in the former case, the irregular *forwent* should surface (similarly to *undergo*∼ *underwent*) and in the latter case it shouldn't (similarly to *tango*∼*tangoed*). Even if speakers treat *forgo* as decomposed, they might remain uncertain about more specific properties of the decomposition that might affect the past tense. In particular, they might be uncertain if and how the *go* in *for+go* relates to simplex *go*.[15] A similar explanation is offered by Nikolaev and Bermel (2022, p. 586) for the defectiveness of compound verbs such as *troubleshoot*. Decompositional uncertainty is also used by İleri and Demirok (2022) for the gap in the 3pl desiderative in Turkish. Regardless of the source of uncertainty, the speaker is not in a position to satisfy KNF with either *forgoed* or *forwent*, and the form is gapped.

5.4 Back to the three worries

With KNF in place, we can now return to the three worries about gaps and doublets: representation, learning, and distribution.

For doublets, the worry was that a specific form (e.g., *dove*) might be expected to block a default (e.g., *dived*). While a solution existed — using hyper-specific versions of *-ed* and ablaut — it did not seem correct. The current proposal addresses this worry by allowing doublets to arise from the existence of multiple grammars within the correct hypothesis. With suitable representations, splitting the grammar for, e.g., *dived/dove* can be much simpler to encode than the hyper-specific entries needed within an individual grammar. For example, the ablaut rule might list $\sqrt{\text{DIVE}}$ as optional within its context while keeping the rest of the grammar intact, a compact way to state two versions of the grammar, one

2007) for discussion and for an argument for a representational framework that allows for partially-ordered constraints rather than a naive multiplication of grammars.

[15] Uncertainty about the connection of a verb to a homophonous irregular verb arises also without decomposition, as in the slang use of *dig*.

with the root in the context and one without.[16] Or the instance of the root in the context of the ablaut rule can be indexed to one or more specific grammars (that can also be indexed by other elements). I will not attempt to investigate the actual representations of multiple grammars here. My point is just that for various representational choices, splitting the grammar will be preferred by a balanced learner to acquiring hyper-specific entries. Both representation and learning of doublets are therefore unproblematic once multiple grammars are allowed within a hypothesis.

The challenges for gaps were more serious, and I return to each of them in turn.

5.4.1 Representation

The worry about representations was that when speakers are asked to provide the missing form in cases of paradigm gaps they often report being uncomfortable, a reaction that is different from familiar cases of unacceptability such as island violations or impossible word orders. The present account offers a handle on this discomfort: gaps arising due to KNF are pragmatic, reflecting an epistemic position that is inadequate for the use of a given form. The unacceptability therefore has an entirely different source from that of island violations and impossible word orders, which reflect ungrammaticality. Pragmatic violations do not necessarily imply discomfort, but for gaps in inflectional morphology such discomfort can be expected: inflectional paradigms are generally complete, so there should be at least one possible form; moreover, adult speakers are supposed to know the inflectional system in their native language; discovering that one does not, in fact, know is embarrassing.

When there is a justification for not knowing, speakers may feel more comfortable with epistemic gaps. Potential cases include children's gaps during early stages of acquisition and adults' gaps outside of their native languages or for forms or paradigms that are sufficiently rare.

Speakers' discomfort with gaps can be expected to be similar to the discomfort of violating KNA, again contrasting with the clear truth-value judgment of an assertion that is known to be false.[17] Differently from KNF, KNA concerns world knowledge, for which it is often perfectly excusable not to know the fact of the matter, and it is generally permissible to say so. But this is not always the case. We are expected to know our own names, for example, and also expected to use them when we introduce ourselves. If I am introducing myself and suddenly realize that I don't know whether my name is Bill or Bob, this will be uncomfortable. (If I strongly suspect it's Bill rather than Bob, introducing myself as

[16] With other instances of this marking, the versions of the grammar will multiply.

[17] There may well be other domains in which epistemic gaps lead to similar discomfort. Certain moral dilemmas might be a case in point, along with more prosaic dilemmas such as how to recycle a given item.

Bill will presumably be better, though still uncomfortable.)

5.4.2 Learning

The worry about learning gaps was that learning the lack of a default requires sufficient indirect negative evidence. Such evidence may support the lexicalization of certain gaps, with potential candidates such as **scissor* and **jink*, but it is an implausible explanation for many other cases, including *forgo*. The current proposal addresses this worry by allowing gaps to arise through speaker uncertainty about the correct hypothesis (while still supporting lexicalized gaps when sufficient indirect negative evidence is available). As in Albright (2003, 2009)'s account, low frequency is perfectly compatible with uncertainty and may in fact contribute to it, so there is a path for gaps to arise even in low-frequency cases.

5.4.3 Distribution

The worry about distribution was that gaps and doublets seemed correlated with the presence of what can informally seem like clashing generalizations but that the correlation was imperfect, as summarized in (2): depending on the particular example we may find no acceptable forms (gaps), a single form (the typical case), or multiple forms (doublets).

On the present account, conflicting generalizations affect the emergence of gaps and doublets indirectly, through learning and its implications for what speakers know about what the correct hypothesis says about a given form. Consider the case of *forgo*, where the potentially clashing generalizations are the default *-ed* and the irregular *go*$\sim$*went*. For *go* itself, competition for specificity within a single grammar leads to *went* being the sole winner, and the same applies when *go* is part of the composition of a complex verb such as *undergo*. Speakers have evidence for the decomposition of *undergo*, both from instances of *underwent* and from the semantic transparency of the decomposition. For verbs that do not decompose, *go*$\sim$*went* is inapplicable and (in the absence of other generalizations) only the default *-ed* applies. Such is the case for *tango* and *embargo*, for which evidence against decomposition comes from instances of *tangoed* and *embargoed* but also from the placement of stress and from the fact that the meaning of these verbs is unrelated to *go*. For *forgo*, speakers can find themselves not knowing what the correct hypothesis says. It is possible (and in fact likely) that the verb decomposes, but the semantic connection to *go* is opaque, and in the absence of evidence from instances of *forwent* in the input, speakers may not reach the bar for knowing that within the correct hypothesis, the verb *forgo* is decomposed and that it does so in a way that ensures the inflection of the verb *go*. Consequently, they will not be in a position to know about either *forgoed* or *forwent* that it is derived by the correct hypothesis. In other words, they cannot use either form without violating KNF, and the past tense of the verb will be a gap.

In other cases, failing to know what the correct hypothesis says about a given form can have nothing to do with decompositional uncertainty. For the masculine singular genitive form in Polish, speakers might simply not know which of the two non-default forms *-a* and *-u* applies. If they hear just one of the forms above noise levels for a given noun, they will know that the correct hypothesis assigns just that ending to that noun. If they hear both, they will have a doublet.[18] But if they hear neither they will be left without knowing and will therefore not be able to satisfy KNF.

6 Further predictions

6.1 Trans-derivational effects

The current proposal explains the appearance of trans-derivational constraints without committing to a grammatical formalism that allows for the statement of actual trans-derivational constraints. Suppose that there is a near-tie between the two leading hypotheses, $H1 = \{G1\}$ and $H2 = \{G2\}$ that conflict regarding a given form X. $G1$ might express X as x_1 while $G2$ expresses it as x_2. In the case of *forgo*, $G1$ might decompose the verb into *for+go* and derive *forwent* while $G2$ might treat it as nondecomposable and derive *forgoed*, with speakers not knowing whether the correct hypothesis is $H1$ or $H2$. On the present account, this would mean that KNF cannot be met for the past tense of *forgo* and that a gap arises.

Suppose, however, that there is separate evidence that speakers can access regarding the decomposition of *forgo*. Such evidence would inform speakers about the choice between $H1$ and $H2$ and allow them to meet KNF. This may look like a trans-derivational constraint connecting the source of evidence about decomposition and the past tense of *forgo*, but it is simply the usual kind of inference about hypotheses that learners carry out, and the grammars themselves need not involve any such constraints. For the English past tense it is not clear to me whether such a source of evidence exists, but as mentioned above, the literature points to such sources for other cases, such as the Spanish and Russian patterns discussed above. The present account explains why such cases can have the appearance of trans-derivational constraints and also why they are generally imperfect and why their shape can vary: they may look like Lexical Conservatism effects in some cases, like directed inference from one specific base to another, and so on.

[18] Given that neither *-a* nor *-u* is a default, the doublet can be represented in a single grammar. As discussed above, for cases such as *dived* and *dove* in English, the grammar will need to be split in order to prevent the irregular *dove* from always blocking the default *dived*.

6.2 Stability of gaps and doublets

The current proposal predicts that the distribution of gaps and doublets will vary to some extent between speakers and over generations. A child who hears *forgoed* a few times, even if this is a production error of speakers for whom this form is gapped, can acquire *forgoed*. If they also happen to hear *forwent* (again, possibly as production errors by speakers for whom the form is gapped) they may acquire a doublet. In this way, gaps can disappear and doublets can appear over time.

It is also possible for gaps to appear and for doublets to disappear over time. For example, a child growing up in an environment that has a doublet such as *dived/dove* might hear just one of the forms above noise level and end up without the grammar split and without the doublet. And if they hear neither form above noise level (which is likely to happen in cases with low frequency) and have no other supporting evidence, they might never reach knowledge about what the correct hypothesis says about the relevant forms. In that case, they will have a gap that was not there for the previous generation. This gap can then entrench itself: the child is not likely to produce the relevant forms, so the lack of knowledge about the correct hypothesis is likely to arise also in the following generation.

6.3 Salvation by uncertainty elimination

The connection proposed here between KNF and KNA suggests a parallel with presuppositions on the view that treats presupposition failure epistemically, as an inability to determine the (possibly classical) truth value of the sentence. This view has been argued for by Fox (2008) and George (2008), as discussed briefly below, and has logical foundations formulated (in different ways) by Kleene (1952) and van Fraassen (1966).

On the epistemic view, *The King of France is bald* has a classical truth value — either true or false — but we cannot tell what it is. Fox (2008) and George (2008) argue that this view makes it possible to predict the pattern of presupposition projection from within complex sentences. Simplifying considerably, presupposition projection is reduced to uncertainty projection. For example, if A is false and the truth value of B is unknown (either true or false but we don't know which), then $A \wedge B$ is false — we can determine this without knowing the truth value of B. $A \vee B$, on the other hand, has an unknown truth value. And if A is true and the truth value of B is unknown, $A \wedge B$ is unknown while $A \vee B$ is true.

A discussion of the intricacies of presupposition projection goes well beyond the scope of the current paper. What I wish to point out is just that, on the current account, we should expect uncertainty elimination to have a counterpart with paradigm gaps. Consider a gap that arises from uncertainty between two hypotheses, $H1$ and $H2$, the former licensing x_1 and not x_2 for a given X and the latter licensing x_2 and not x_1. In a context in which speakers do not need to actually use the form, KNF is not at stake for X, and the sentence can be acceptable. Any process that obliterates the difference between x_1 and x_2 should in principle serve this purpose. An obvious process of this kind is phonological deletion, and building on observations by Abels (2019), Adamson (2019), and Mendes and Nevins (2022), I believe that the prediction of the current account is correct.

If a structure includes the past participle of *stride* but is phonologically null (for example, due to VP ellipsis), then speakers might feel comfortable: while they might not know whether the correct hypothesis licenses *strided* or *stridden*, they might still know that it licenses some form, and whichever form it is, the phonological output in an ellipsis context will sound the same (namely, null). In such a case, KNF is vacuously satisfied, and speakers might be willing to accept the relevant form. As Adamson (2019) notes, this is the case: *Jane strode into the room, even though she shouldn't have*, where the problematic participial form of *stride* is inside the elided VP, is judged as acceptable (p. 58).[19,20]

6.4 Beyond inflectional morphology?

The present account is stated in terms of general principles of grammar and cognition. Why, then, does the literature on gaps seem to focus specifically on (inflectional) morphology? To some extent, this focus makes sense. Morphological processes vary between items, so for a given form, speakers can easily be left with uncertainty. This is especially so for inflectional morphology, where speakers can generally assume (a) that a form exists, and (b) that they should know what this form is. So when they are uncertain, their knowledge is incomplete, which on the current account is how gaps arise.

In derivational morphology, it is often the case that no form exists. E.g., *-ess* in English, which attaches to several nouns, such as *lion*, but not to others, such as *cat*. See Embick

[19]As pointed out to me by Jonathan Bobaljik (p.c.), uncertainty can in principle be eliminated through other means as well. In particular, he notes that certain cases in the syntactic literature that have been argued to require matching but can be rescued by syncretism might be reanalyzed in terms of uncertainty and its elimination. (Sims 2015, 29–30 considers such cases briefly in the context of paradigmatic gaps but draws a distinction between them and gaps in inflectional paradigms.)

[20]The works cited treat rescue by deletion differently from the current account. All three propose that gaps that are due to ungrammaticality at PF are unproblematic when, as in ellipsis, phonological realization is not required. Mendes and Nevins (2022) support this claim with evidence that gaps whose source seems to be semantic rather than morpho-phonological cannot be rescued by ellipsis. As far as I can tell, this is compatible with the current account: semantic gaps are problematic regardless of whether they also involve a KNF violation, so phonological deletion should not rescue them.

I will not attempt to settle the question of whether deletion repairs PF ill-formedness in general. On the current account, and as far as paradigm gaps are concerned, it is not required to do so: deletion simply makes it possible to avoid a pragmatic problem (namely, an inability to satisfy KNF) and is not required to repair ungrammaticality.

and Marantz (2008) for discussion. Speakers can know such facts, which makes **catess* ungrammatical (since speakers know that it cannot be derived in the correct hypothesis) rather than epistemically gapped. Similarly for semantic gaps such as those analyzed by Harley (2014) and Mendes and Nevins (2022). For example, speakers presumably know that the correct hypothesis has no singular form for *scissors* and no context-independent meaning for *cran-* or *jink*. Violations for these cases are ungrammatical rather than uncomfortable.

Phonology, syntax, and compositional semantics, on the other hand, tend to apply across-the-board. There will therefore be no occasion for two grammars to vary in whether, e.g., a given determiner composes with a given noun or whether topicalization can apply to the resulting noun phrase.

But there are places where uncertainty about items (perhaps because of morphological considerations) might give rise to uncertainty that seems to go beyond morphology. I will briefly mention two potential cases of this kind but will not attempt to analyze them in any detail within the present paper.

One example, pointed out to me by Jonathan Bobaljik (p.c.), concerns the possessive form of coordination with personal pronouns, discussed by Herriman (1999) and Payne (2011). For at least some English speakers, $^{??}$*Ann and my paper*, $^{??}$*Ann's and my paper*, and $^{??}$*Ann and me's paper* are all degraded, with no acceptable way to form the relevant possessive. This might be amenable to an analysis in terms of uncertainty: speakers might not know what the correct hypothesis says for such cases.

Another potential example, pointed out to me by Nina Haslinger (p.c.) and discussed by Haider (1993), Koopman (1995), Fanselow and Féry (2002), Vikner (2005), and others concerns certain complex verbs in German and Dutch. In both languages, some complex verbs split in contexts where the verb needs to move to the V2 position while others don't. For example, German *aufstehen* 'to get up' splits, as in *Er steht nicht auf* "He doesn't get up", with the prefix *auf* 'up' staying in its final position and the stem *steht* 'stand' moving to the V2 position. The split and the movement are both obligatory: **Er aufsteht nicht* (with an attempted movement to V2 of both the prefix and the stem) and **Er nicht aufsteht* (with neither affix nor stem moving) are both bad. German *verstehen* 'to understand', on the other hand, does not split and moves to the V2 position as a whole, as in *Er versteht nicht* "He doesn't understand". Movement is again obligatory, but here no split is possible: **Er nicht versteht* and **Er steht nicht ver* are ungrammatical. For certain complex verbs, however, such as German *uraufführen* 'to premiere', *voranmelden* 'to preregister', and *bergsteigen* 'to mountain-climb', speakers have a gap in V2 contexts: **Wir uraufführten das Stück* "We premiered the play" (with the verb *führen* moving along with its two prefixes *ur-* and *auf-* to V2) is unacceptable, as are **Wir aufführten das Stück ur* (with *ur-* stranded and the rest moving to V2) and **Wir führten das Stück urauf* (with *ur-auf-* stranded and just the verb moving to V2). Such verbs are otherwise unremarkable, and when they do not need to move to the V2 position they cause no problems, as in the acceptable *Wir wollen das Stück uraufführen* “We want to premiere the play” (where the modal *wollen* ‘want.to’ occupies the V2 position and the complex *uraufführen* stays in place, so the issue of splitting does not arise).

One can encode the immobility of verbs like *uraufführen* directly within the grammar, as Vikner (2005) does. On his account, the relevant verbs need to satisfy the conflicting demands of splitting and non-splitting verbs. It is unclear why this should be or why children should learn this, especially in the case of rare or nonce forms. It is also worth highlighting that the problematic verbs generally pattern with splitting verbs for purposes of infixation. For example, they allow the infixation of *zu* ‘to’, as in *uraufzuführen* and *bergzusteigen*, similarly to the splitting *aufstehen* (*aufzustehen* vs. **zu aufstehen*) and differently from the non-splitting *verstehen* (**verzustehen* vs. *zu verstehen*). It strikes me as at least preliminarily plausible that the problem is that speakers are faced with conflicting evidence: they have strong evidence that the verbs split but also good reason to think that the prefix cannot be stranded. If so, they are left without knowing about either of the possibilities that it is licensed by the correct hypothesis.

The two cases above are suggestive of uncertainty gaps beyond the strictly morphological domain, a phenomenon that is predicted by the current account but not, as far as I can see, for the accounts surveyed in section 4, all of which tie gaps to specific properties of morphological paradigms. Of course, a proper analysis would require more than the brief overview provided here.

7 Conclusion

I conclude that at this point gaps and doublets teach us very little about morphology. Doublets raised a potential challenge, mostly due to the apparent need for hyper-specific entries within morphological frameworks that involve competition for specificity, possibly motivating the abandonment of competition for specificity, as is done in OT. But the possibility of encoding multiple grammars within a hypothesis eliminated that challenge. The challenge raised by the representation, learning, and distribution of gaps seemed more serious, possibly motivating the rather major departures from standard assumptions in morphology entertained in the literature. But with uncertainty about what the correct hypothesis says, the properties of gaps become considerably less puzzling.

My account of gaps relied also on KNF, which is not part of existing proposals. Still, KNF seems no more stipulative than other choices of how to handle uncertainty. In light of the connection to KNA, the present choice may in fact be more natural than some other prominent possibilities, as discussed above, and I tried to show that it also provides a better handle on the empirical pattern of gaps than existing accounts.

References

Abels, Klaus. 2019. On "sluicing" with apparent massive pied-piping. *Natural Language & Linguistic Theory* 37:1205–1271.

Adamson, Luke James. 2019. Derivational trapping and the morphosyntax of inflectionlessness. Doctoral Dissertation, University of Pennsylvania.

Albright, Adam. 2003. A quantitative study of Spanish paradigm gaps. In *Proceedings of the 22nd West Coast Conference on Formal Linguistics*, 1–14.

Albright, Adam. 2009. Lexical and morphological conditioning of paradigm gaps. In *Modeling ungrammaticality in Optimality Theory*, ed. Sylvia Blaho and Curt Rice, 117–164. Equinox Publishing.

Albright, Adam, and Bruce Hayes. 2002. Modeling English past tense intuitions with minimal generalization. In *Proceedings of the ACL-02 workshop on Morphological and phonological learning*, volume 6, 58–69. Association for Computational Linguistics.

Albright, Adam, and Bruce Hayes. 2003. Rules vs. analogy in English past tenses: a computational/experimental study. *Cognition* 90:119–161.

Andrews, Avery D. 1990. Unification and morphological blocking. *Natural Language and Linguistic Theory* 8:507–557.

Anttila, Arto. 2004. Variation and phonological theory. *The handbook of language variation and change* 206–243.

Anttila, Arto. 2007. Variation and optionality. *The Cambridge handbook of phonology* 519–536.

Arregi, Karlos, and Andrew Nevins. 2014. A monoradical approach to some cases of disuppletion. *Theoretical Linguistics* 40:311–330.

Baerman, Matthew. 2008. Historical observations on defectiveness: the first singular non-past. *Russian Linguistics* 32:81–97.

Baerman, Matthew. 2011. Defectiveness and homophony avoidance. *Journal of Linguistics* 47:1–29.

Benton, Matthew A. 2024. Knowledge is the norm of assertion. In *Contemporary debates in epistemology*, ed. Blake Roeber, Ernest Sosa, Matthias Steup, and John Turri, 329–339. Wiley-Blackwell, 3 edition.

Berko, Jean. 1958. The child's learning of English morphology. *Word* 14:150–177.

Bobaljik, Jonathan David. 2017. Distributed morphology. In *Oxford research encyclopedia of linguistics*. Oxford University Press.

Boyé, Gilles, and Patricia Cabredo Hofherr. 2010. Defectivity as stem suppletion in French and Spanish verbs. In *Defective paradigms: Missing forms and what they tell us*, ed. Matthew Baerman, Greville G. Corbett, and Dunstan Brown, 63–92. Oxford: Oxford University Press.

Bybee, Joan L., and Carol Lynn Moder. 1983. Morphological classes as natural categories. *Language* 59:251–270.

Caha, Pavel. 2009. The nanosyntax of case. Doctoral Dissertation, University of Tromsø.

Caha, Pavel. forthcoming. Nanosyntax: Some key features. In *The cambridge handbook of distributed morphology*, ed. Artemis Alexiadou, Ruth Kramer, Alec Marantz, and Isabel Oltra-Massuet. Cambridge: Cambridge University Press.

Caha, Pavel, Karen De Clercq, and Guido Vanden Wyngaerd, ed. 2025. *Nanosyntax and the lexicalization algorithm*. Oxford: Oxford University Press.

Calamaro, Shira, and Gaja Jarosz. 2015. Learning general phonological rules from distributional information: A computational model. *Cognitive Science* 39:647–666.

Chomsky, Noam. 1957. *Syntactic structures*. The Hague: Mouton.

Chomsky, Noam. 1965. *Aspects of the theory of syntax*. Cambridge, MA: MIT Press.

Chomsky, Noam, and Morris Halle. 1968. *The sound pattern of English*. New York: Harper and Row Publishers.

Dąbrowska, Ewa. 2001. Learning a morphological system without a default: the Polish genitive. *Journal of child language* 28:545–574.

Daland, Robert, Andrea D. Sims, and Janet Pierrehumbert. 2007. Much ado about nothing: A social network model of Russian paradigmatic gaps. In *Proceedings of the 45th annual meeting of the Association of Computational Linguistics*, 936–943.

Dell, François. 1981. On the learnability of optional phonological rules. *Linguistic Inquiry* 12:31–37.

DeRose, Keith. 2002. Assertion, knowledge, and context. *The Philosophical Review* 111:167–203.

Douven, Igor. 2006. Assertion, knowledge, and rational credibility. *The Philosophical Review* 115:449–485.

van Elswyk, Peter, and Matthew A. Benton. 2023. Assertion remains strong. *Philosophical Studies* 180:27–50.

Embick, David, Johanna Benz, and Lefteris Paparounas. 2023. Blocking effects. In *The wiley blackwell companion to morphology*, ed. Peter Ackema, Sabrina Bendjaballah, Eulàlia Bonet, and Antonio Fábregas, 271–312. Wiley-Blackwell.

Embick, David, and Alec Marantz. 2008. Architecture and blocking. *Linguistic Inquiry* 39:1–53.

Fanselow, Gisbert, and Caroline Féry. 2002. Ineffability in grammar. In *Resolving conflicts in grammars*, 265–307. Buske Verlag.

Fox, Danny. 2008. Two short notes on Schlenker's theory of presupposition projection. *Theoretical Linguistics* 34:237–252.

van Fraassen, Bas. 1966. Singular terms, truth-value gaps, and free logic. *Journal of Philosophy* 63:481–495.

George, Benjamin R. 2008. A new predictive theory of presupposition projection. In *Proceedings of SALT 18*, 358–375.

Goldsmith, John. 2001. Unsupervised learning of the morphology of a natural language. *Computational Linguistics* 27:153–198.

Goldwater, Sharon, and Mark Johnson. 2004. Priors in Bayesian learning of phonological rules. In *7th Annual Meeting of the ACL Special Interest Group on Computational Phonology*, 35–42.

Gorman, Kyle, and Charles Yang. 2019. When nobody wins. In *Competition in inflection and word-formation*, ed. Franz Rainer, Francesco Gardani, Wolfgang U. Dressler, and Hans Christian Luschützky, 169–193. Cham: Springer.

Graham, Peter J., and Nikolaj J. L. L. Pedersen. 2024. Knowledge is not our norm of assertion. In *Contemporary debates in epistemology*, ed. Blake Roeber, Ernest Sosa, Matthias Steup, and John Turri. Wiley-Blackwell, 3 edition.

Haider, Hubert. 1993. *Deutsche Syntax, Generativ: Vorstudien zur Theorie einer projektiven Grammatik*, volume 325. Gunter Narr Verlag.

Halle, Morris. 1973. Prolegomena to a theory of word formation. *Linguistic Inquiry* 4:3–16.

Halle, Morris. 1997. Distributed Morphology: Impoverishment and fission. In *Papers at the interface*, ed. B. Bruening, Y. Kang, and M. McGinnis, 125–149. Cambridge, Mass.: MIT Working Papers in Linguistics.

Halle, Morris, and Alec Marantz. 1993. Distributed morphology and the pieces of inflection. In *The view from Building 20*, ed. Kenneth Hale and Jay Keyser, 111–176. Cambridge, MA: MIT Press.

Halle, Morris, and Alec Marantz. 1994. Some key features of distributed morphology. In *Papers on phonology and morphology*, 275–288. MIT Working Papers in Linguistics.

Harley, Heidi. 2014. On the identity of roots. *Theoretical Linguistics* 40:225–276.

Harris, James Wesley. 1967. Spanish phonology. Doctoral Dissertation, Massachusetts Institute of Technology.

Herriman, Jennifer. 1999. Coordinating genitives in determinative function in English. *Studia Neophilologica* 71:1–12.

Hetzron, Robert. 1975. Where the grammar fails. *Language* 51:859–872.

Hudson, Richard. 2000. I amn't. *Language* 76:297–323.

İleri, Muhammed, and Ömer Demirok. 2022. A paradigm gap in Turkish. In *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic*, volume 7, 1–15.

Katzir, Roni. 2014. A cognitively plausible model for grammar induction. *Journal of Language Modelling* 2:213–248.

Kleene, Stephen Cole. 1952. *Introduction to metamathematics*. Amsterdam: North-Holland.

Koopman, Hilda. 1995. On verbs that fail to undergo V-second. *Linguistic Inquiry* 26:137–164.

Kottum, Steinar E. 1981. The genitive singular form of masculine nouns in Polish. *Scando-Slavica* 27:179–186.

Kroch, Anthony S. 1994. Morphosyntactic variation. In *Papers from the 30th Regional Meeting of the Chicago Linguistic Society*, ed. Katharine et al. Beals.

Lackey, Jennifer. 2007. Norms of assertion. *Noûs* 41:594–626.

Maiden, Martin, and Paul O'Neill. 2010. On morphomic defectiveness: Evidence from the Romance languages of the Iberian peninsula. In *Defective paradigms: Missing forms and what they tell us*, ed. Matthew Baerman, Greville G. Corbett, and Dunstan Brown.

Mandelkern, Matthew, and Kevin Dorst. 2022. Assertion is weak. *Philosophers' Imprint* 22.

Manzini, M. Rita, and Kenneth Wexler. 1987. Parameters, binding theory, and learnability. *Linguistic Inquiry* 18:413–444.

de Marcken, Carl. 1996. Unsupervised language acquisition. Doctoral Dissertation, MIT, Cambridge, MA.

Marcus, Gary F. 1993. Negative evidence in language acquisition. *Cognition* 46:53–85.

McCarthy, John. 2000. Harmonic serialism and parallelism. In *Proceedings of the North East Linguistic Society*, ed. Masako Hirotani, Andries Coetzee, Nancy Hall, and Ji-yung Kim, 501–524. Rutgers University: Graduate Linguistic Student Association.

McCarthy, John J. 2005. Optimal paradigms. In *Paradigms in phonological theory*, ed. Laura J. Downing, T. A. Hall, and Renate Raffelsiefen, 170–210. Oxford University Press.

McCarthy, John J. 2010. An introduction to harmonic serialism. *Language and Linguistics Compass* 4:1001–1018.

Mendes, Gesoel, and Andrew Nevins. 2022. When ellipsis can save defectiveness and when it can't. *Linguistic Inquiry* 54:182–196.

Müller, Gereon. 2020. *Inflectional morphology in harmonic serialism*. Equinox Publishing.

Nevins, Andrew. 2014. Book review:"Defective paradigms: Missing forms and what they tell us". *Revista Linguística* 10.

Nikolaev, Alexandre, and Neil Bermel. 2022. Explaining uncertainty and defectivity of inflectional paradigms. *Cognitive Linguistics* 33:585–621.

O'Donnell, Timothy J. 2015. *Productivity and reuse in language: A theory of linguistic*

computation and storage. MIT Press.
Orgun, Cemil Orhan, and Ronald L. Sprouse. 1999. From mparse to control: deriving ungrammaticality. *Phonology* 16:191–224.
Payne, John. 2011. Genitive coordinations with personal pronouns. *English Language and Linguistics* 15:363–385.
Pertsova, Katya. 2005. How lexical conservatism can lead to paradigm gaps. *UCLA working papers in phonology* 11.
Pertsova, Katya. 2016. Transderivational relations and paradigm gaps in Russian verbs. *Glossa: a journal of general linguistics* 1.
Pertsova, Katya. forthcoming. Inflectional defectiveness (through the lens of acquisition of morphophonological alternations in Russian). To appear in the Blackwell Companion to Phonology.
Pinker, Steven. 1998. Words and rules. *Lingua* 106:219–242.
Prasada, Sandeep, and Steven Pinker. 1993. Generalisation of regular and irregular morphological patterns. *Language and Cognitive Processes* 8:1–56.
Prince, Alan, and Paul Smolensky. 1993. Optimality theory: Constraint interaction in generative grammar. Technical report, Rutgers University, Center for Cognitive Science.
Raffelsiefen, Renate. 2004. Absolute ill-formedness and other morphophonological effects. *Phonology* 21:91–142.
Rasin, Ezer, Iddo Berger, Nur Lan, Itamar Shefi, and Roni Katzir. 2021. Approaching explanatory adequacy in phonology using Minimum Description Length. *Journal of Language Modelling* 9:17–66.
Rasin, Ezer, and Roni Katzir. 2016. On evaluation metrics in Optimality Theory. *Linguistic Inquiry* 47:235–282.
Rice, Curt. 2007. Gaps and repairs at the phonology-morphology interface. *Journal of Linguistics* 43:197–221.
Rissanen, Jorma. 1978. Modeling by shortest data description. *Automatica* 14:465–471.
Roeper, Thomas. 1999. Universal bilingualism. *Bilingualism: Language and Cognition* 2:169–186.
Sims, Andrea D. 2006. Minding the gaps: Inflectional defectiveness in a paradigmatic theory. Doctoral Dissertation, The Ohio State University.
Sims, Andrea D. 2009. Why defective paradigms are, and aren't, the result of competing morphological patterns. In *Proceedings from the Annual Meeting of the Chicago Linguistic Society*, ed. Malcolm Elliott, James Kirby, Osamu Sawada, Eleni Staraki, and Suwon Yoon, volume 43, 267–281. Chicago Linguistic Society.
Sims, Andrea D. 2015. *Inflectional defectiveness*. Cambridge Studies in Linguistics. Cambridge University Press.
Sims, Andrea D. 2017. Slavic morphology: Recent approaches to classic problems, illus-

trated with Russian. *Journal of Slavic Linguistics* 25:489–524.

Sims, Andrea D. 2023. Defectiveness. In *The Wiley Blackwell companion to morphology*, ed. Peter Ackema, Sabrina Bendjaballah, Eulàlia Bonet, and Antonio Fábregas, 573–606. Wiley-Blackwell.

Slioussar, Natalia, and Maria Kholodilova. 2013. Paradigm leveling in non-standard Russian. In *Formal approaches to Slavic linguistics: The second MIT meeting 2011*, ed. Alexander Podobryaev, 243–258. Ann Arbor, Michigan: Michigan University Press.

Solomonoff, Ray J. 1964. A formal theory of inductive inference, parts I and II. *Information and Control* 7:1–22, 224–254.

Starke, Michal. 2009. Nanosyntax: A short primer to a new approach to language. *Nordlyd* 36:1–6.

Steriade, Donca. 1997. Lexical conservatism. In *Linguistics in the morning calm*, 157–179. Linguistic Society of Korea Hanshin Publishing House.

Tomas, Ekaterina, Ruben van de Vijver, Katherine Demuth, and Peter Petocz. 2017. Acquisition of nominal morphophonological alternations in Russian. *First Language* 37:453–474.

Turri, John. 2016. *Knowledge and the norm of assertion: An essay in philosophical science*. Open Book Publishers.

Unger, Peter. 1975. *Ignorance: A case for scepticism*. OUP Oxford.

Vikner, Sten. 2005. Immobile complex verbs in Germanic. *The Journal of Comparative Germanic Linguistics* 8:83–115.

Williamson, Timothy. 1996. Knowing and asserting. *The Philosophical Review* 105:489–523.

Williamson, Timothy. 2000. *Knowledge and its limits*. Oxford: Oxford University Press.

Yang, Charles. 2000. Knowledge and learning in natural language. Doctoral Dissertation, MIT, Cambridge, MA.

Yang, Charles. 2005. On productivity. *Yearbook of Language Variation* 5:333–370.

Yang, Charles. 2016. *The price of linguistic productivity: How children learn to break the rules of language*. MIT Press.